3. Data Flow & System Behavior

3.1 Data Flow Diagrams (DFD)

The Data Flow Diagrams describe how data moves through the Medicare system: where it comes from, which processes transform it, and where it is stored. The diagrams use the Yourdon / DeMarco notation and are presented in three levels, each one balanced with the level above it.

Symbol	Meaning
Rectangle	External entity (Patient, Doctor, Admin, Payment Gateway)
Circle	Process that transforms data
Open-ended rectangle	Data store (D1 - D5)
Arrow	Data flow, labeled with the data it carries

3.1.1 Level 0 - Context Diagram

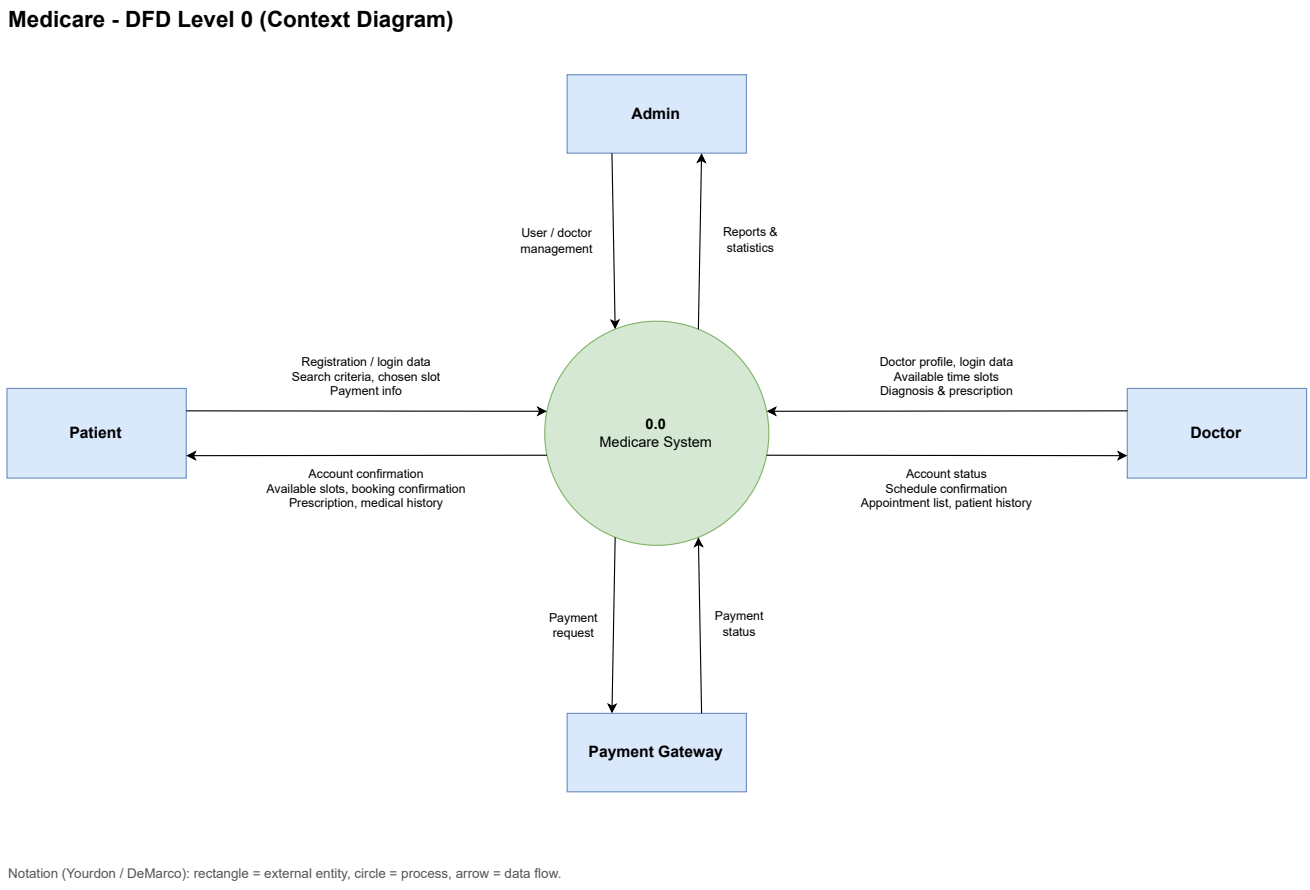


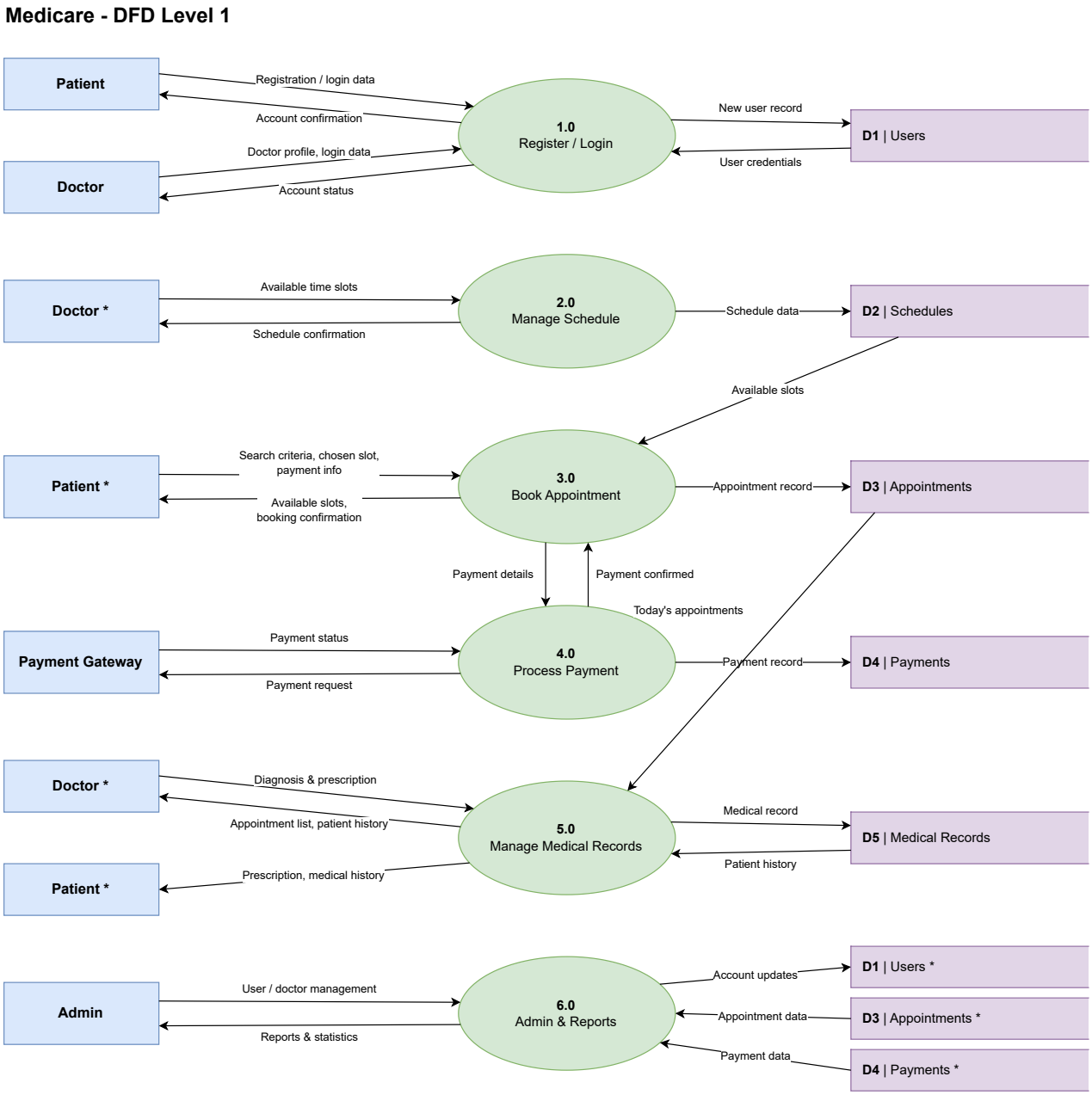
Figure 3.1: Medicare DFD Level 0 (Context Diagram)

The context diagram shows the whole Medicare system as a single process (0.0) and defines the system boundary. Four external entities interact with it:

- **Patient:** registers, searches for doctors, books appointments and pays; receives booking confirmations, prescriptions and medical history.

- **Doctor:** provides profile data, available time slots, diagnoses and prescriptions; receives the appointment list and patient history.
- **Admin:** manages users and doctors; receives reports and statistics.
- **Payment Gateway:** receives payment requests and returns the payment status.

3.1.2 Level 1 - Detailed DFD



* Entities and data stores marked with * are duplicates, drawn again only to keep the lines short.

Figure 3.2: Medicare DFD Level 1

Level 1 decomposes process 0.0 into six main processes and five data stores:

Process	Description	Data store
1.0 Register / Login	Registers patients and doctors and verifies their login credentials.	D1 Users
2.0 Manage Schedule	Doctors define their available time slots.	D2 Schedules
3.0 Book Appointment	Patients search available slots and book an appointment, then the booking is sent for payment.	D2, D3 Appointments
4.0 Process Payment	Sends the payment request to the payment gateway and records the result.	D4 Payments
5.0 Manage Medical Records	Doctors record diagnoses and prescriptions; patients view their prescriptions and history.	D3, D5 Medical Records
6.0 Admin & Reports	Admin manages users and doctors and generates reports.	D1, D3, D4

*Note: entities and data stores marked with * are duplicates, drawn again only to keep the diagram readable.*

The diagram is balanced with Level 0: every data flow between an external entity and the system in the context diagram appears in Level 1, distributed over the processes that handle it.

3.1.3 Level 2 - Process 3.0 Book Appointment

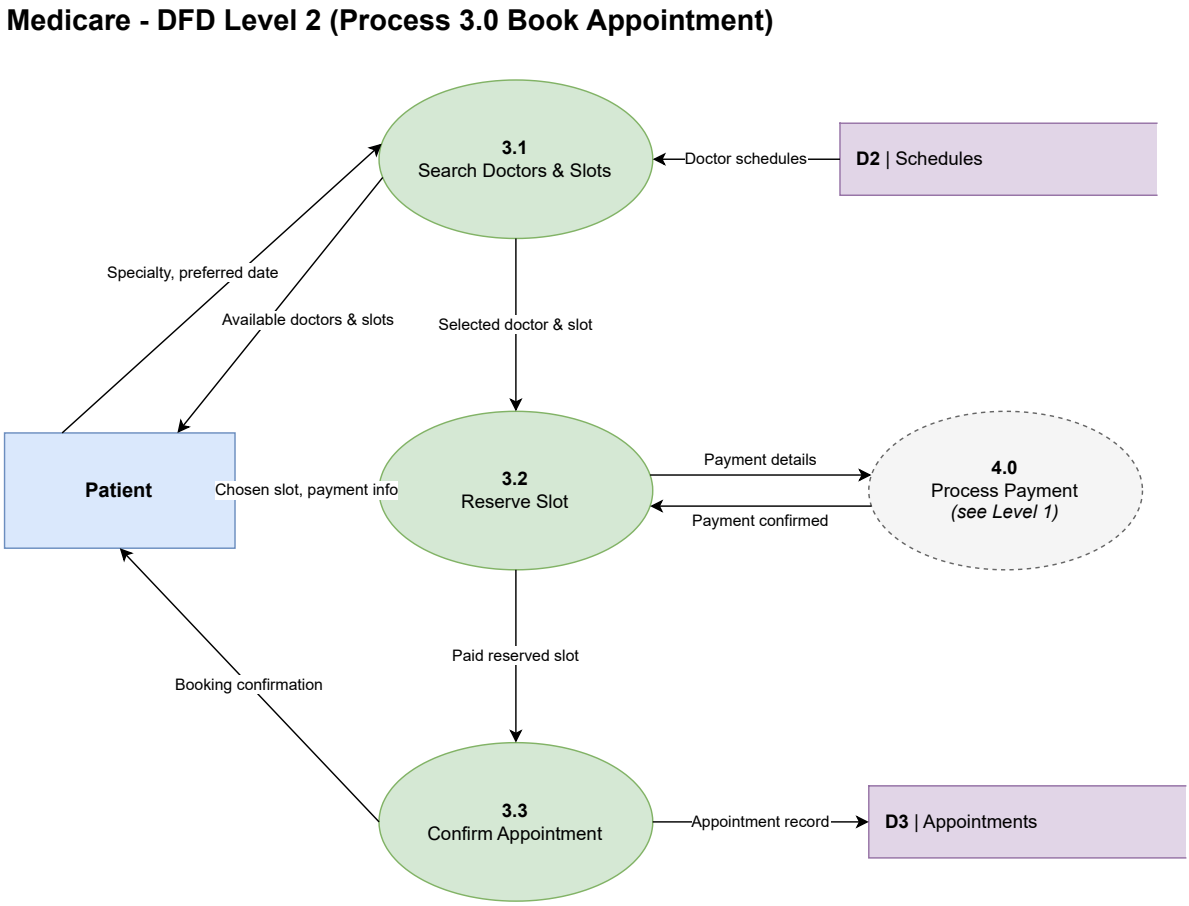


Figure 3.3: Medicare DFD Level 2 for process 3.0 Book Appointment

- **3.1 Search Doctors & Slots:** the patient enters the specialty and preferred date; the process reads doctor schedules from D2 and returns the available doctors and slots.
- **3.2 Reserve Slot:** the patient chooses a slot and enters payment info; the process sends the payment details to process 4.0 and waits for the payment confirmation.
- **3.3 Confirm Appointment:** the paid slot is saved as an appointment record in D3 and a booking confirmation is sent to the patient.

Level 2 is balanced with process 3.0 in Level 1: it has the same inputs and outputs (Patient, D2, D3 and process 4.0).